

NETWORKED CONTROL

Prof. Marcello Farina

NAME/FAMILY NAME..... SOLUTIONS.....
PERSONAL CODE.....
SIGNATURE.....
DATE..... 12/01/'22.....

Notes:

- The solutions must be written in the pages of this leaflet.
- At the end of the exam only this file is accepted, **without any additional page**.
- No notes or books are accepted, neither in paper nor electronic form.
- Clarity and precision will be matter of evaluation.

$$\begin{aligned}\dot{x} &= \log_e 2 \cdot x + \log_e 4 \cdot u \\ &= \log_e 2 \cdot x + 2 \log_e 2 \cdot u\end{aligned}$$

A. Eigenvalue: $\lambda = \log_e 2 = 0.6931 > 0 \Rightarrow$ Unstable system
 Mode: $e^{\lambda t} = e^{\log_e 2 \cdot t} = 2^t$

B. 2) According to the theory, $\frac{N_B}{h} \geq \lambda \log_2 e = \log_e 2 \log_2 e$
 $= \log_2 e^{\log_e 2} = \log_2 2 = 1$

b) $h \leq N_B \cdot \lambda \log_e e = N_B = 2 \quad h_{\max} = 2 \rightarrow$

c) $h = 1 \rightarrow, N_B = 2, N = 2^{N_B} = 4, u_{\min} = -1, u_{\max} = 1$

$$x_{k+1} = f x_k + g u_k \quad f = e^{ah} = 2$$

$$g = \frac{b}{a} (e^{ah} - 1) = 2$$

$$x_{k+1} = 2x_k + 2u_k$$

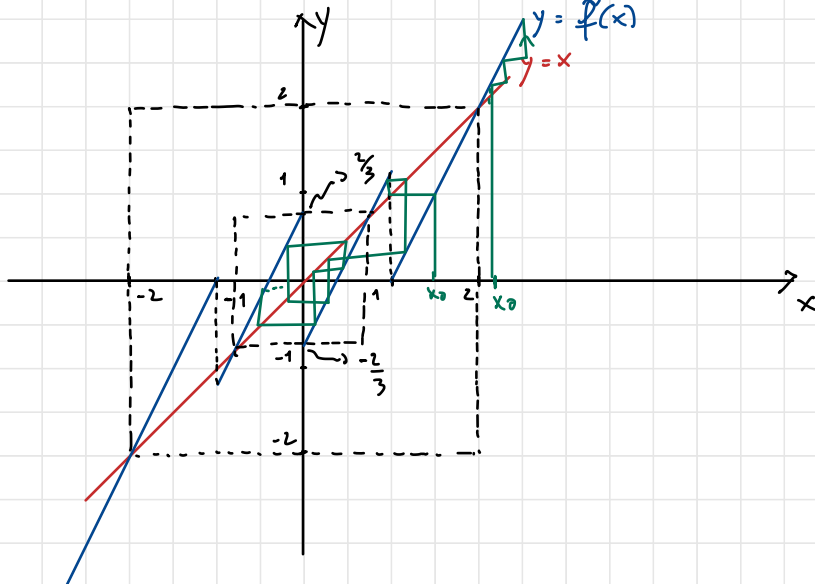
$$\bar{u}_i = u_{\max} - \rho \frac{u_{\max} - u_{\min}}{N-1} = 1 - \frac{2\rho}{3}$$

$$\begin{aligned}\bar{x}_i &= \frac{b}{a} \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) \\ &= 2 \left(-1 + \frac{i}{2} \right)\end{aligned}$$

$$\begin{aligned}\bar{u}_0 &= 1, \bar{u}_1 = \frac{1}{3}, \bar{u}_2 = -\frac{1}{3}, \bar{u}_3 = -1 \\ \bar{x}_0 &= -2, \bar{x}_1 = -1, \bar{x}_2 = 0, \bar{x}_3 = 1, \bar{x}_4 = 2\end{aligned}$$

$$u_k = \begin{cases} 1 & \text{for } x_k \leq -1 \\ \frac{1}{3} & \text{for } -1 < x_k \leq 0 \\ -\frac{1}{3} & \text{for } 0 < x_k \leq 1 \\ -1 & \text{for } x_k > 1 \end{cases}$$

$$x_{k+1} = \hat{f}(x_k) = \begin{cases} 2x_k + 2 & \text{for } x_k \leq -1 \\ 2x_k + \frac{2}{3} & \text{for } x_k \in [-1, 0] \\ 2x_k - \frac{2}{3} & \text{for } x_k \in (0, 1] \\ 2x_k - 2 & \text{for } x_k > 1 \end{cases}$$



IF $|x_0| = 2 \Rightarrow |x_k| = 2 \quad \forall k$

$|x_0| = \frac{2}{3} \Rightarrow |x_k| = \frac{2}{3} \quad \forall k$

$|x_0| > 2 \Rightarrow |x_k| \rightarrow +\infty \text{ as } k \rightarrow +\infty$

$|x_0| \in (\frac{2}{3}, 2) \Rightarrow x_k \rightarrow (-\frac{2}{3}, \frac{2}{3}) \text{ as } k \rightarrow +\infty$

$|x_0| < \frac{2}{3} \Rightarrow x_k \in (-\frac{2}{3}, \frac{2}{3}) \quad \forall k$

NOTE THAT BOTH $[-\frac{2}{3}, \frac{2}{3}]$ AND $[-2, 2]$ ARE INVARIANT SET, FROM THE DEFINITION!

d) No, it is not unique, since $\underline{N_B} < \log_2 e$ and we are not operating at the rate limit.
(SEE THEORY FOR MORE DETAILS)

FOR QUESTIONS 2) AND 3) SEE THEORY,